



SPACE DESIGN LOAD SUMMARY

Studio 1_0

Level: Level 0 · Internal condition: TM59_Bathroom

Test Project
Project no. P-001
Prepared by SAM Tests
2026-09-25

DESIGN CRITERIA

	Heating	Cooling
Room set point	16.0 °C	26.0 °C
Outdoor dry bulb	-4.9 °C	28.4 °C
Outdoor RH (coincident)	90 %	40 %

ROOM HUMIDITY

Humidification set point	—
lower RH limit	
Dehumidification set point	—
upper RH limit	

SIZING (TAS DESIGN LOADS)

	Heating	Cooling
Design load	1,368 W	0 W
Design load per area	263.1 W/m ²	0.0 W/m ²
Sizing multiplier	not set	not set

Design loads: from Tas sizing — date/currency not recorded

HEATING

Design day	Leeds_TRY ANN HTG 100% CONDS DB	
	Design day	Full year
Peak load	2,268 W	802 W
Peak hour	15:00–16:00	1 Jan 00:00–01:00
Room dry bulb	15.1 °C	13.4 °C
Room resultant	13.2 °C	12.6 °C
Room RH	21 %	84 %
Room humidity ratio	2.2 g/kg	8.1 g/kg
Outdoor dry bulb	—	3.3 °C
Outdoor RH	—	100 %

SENSIBLE COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (sensible)	0	105
Equipment (sensible)	0	85
Infiltration / ventilation	-321	-154
Building heat transfer	-664	-327
External conduction — opaque	-895	-328
External conduction — glazing	-388	-184

LATENT COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (latent)	0	77

Zero at both peaks (not listed): Solar, Lighting, Air movement (inter-zone), Air handling unit, Equipment (latent)
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

COOLING

Design day	Leeds_TRY ANN CLG 0% CONDS DB=>GRad	
	Design day	Full year
Peak load	1,973 W	1,972 W
Peak hour	00:00–01:00	3 Jul 19:00–20:00
Room dry bulb	19.8 °C	19.0 °C
Room resultant	21.5 °C	20.6 °C
Room RH	100 %	58 %
Room humidity ratio	14.5 g/kg	8.0 g/kg
Outdoor dry bulb	—	17.5 °C
Outdoor RH	—	59 %

SENSIBLE COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Solar	0	836
Lighting	0	150
Occupancy (sensible)	105	150
Equipment (sensible)	85	450
Infiltration / ventilation	-9	-53
Building heat transfer	971	298
External conduction — opaque	898	219
External conduction — glazing	-77	-78

LATENT COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (latent)	77	110

Zero at both peaks (not listed): Air movement (inter-zone), Air handling unit, Equipment (latent)
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

RESULTS

	Heating	Cooling
Status	Available	Available
Result source	Tas	Tas
Read into model	27 Aug 2026 18:38	27 Aug 2026 18:38
Matches current model	not recorded	not recorded

Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year.
Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied.